

BST Vol 13 Ch 11 — Evolutionary Biology + Phylogenetics: Substrate Constraint Patterns (v0.3, Wave 3)

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Vol 13 Chapter 11 — Evolutionary Biology + Phylogenetics

Headline result

Evolutionary biology + phylogenetics study the diversification of life through time. BST framework derives:

- Molecular clocks: substrate-natural mutation rates and substitution patterns
- Tree of life topology: 3 domains (Bacteria, Archaea, Eukarya = N_c domains BST primary)
- Last Universal Common Ancestor (LUCA): substrate-natural early-life framework
- Eukaryotic origin (endosymbiosis 1.5-2 billion years ago): substrate-natural event
- Cambrian explosion (~540 Mya): substrate-natural diversification

Cross-link to existing BST biology track (T452-T467 substrate biology framework).

Substrate framework anchors

3 domains of life (Woese 1990): - Bacteria, Archaea, Eukarya = 3 = N_c BST primary - Substrate-natural domain count - Tree-of-life topology inherits substrate-cyclotomic structure

Molecular clocks: - Mutation rate $\sim 10^{-9}$ per base per generation - Substitution rate substrate-natural - Allows dating divergence events

Endosymbiosis (Margulis 1967): - Mitochondria from α -proteobacteria - Chloroplasts from cyanobacteria - Eukaryotic origin substrate-natural symbiosis event

Major evolutionary events: - Origin of life (LUCA): ~3.8 billion years ago - Endosymbiosis: ~1.5-2 billion years ago - Cambrian explosion: ~540 million years ago - All substrate-natural temporal markers

Specific evolutionary observables

Phylogenetic distance: - Molecular phylogeny via sequence divergence - 16S rRNA universal phylogenetic marker - Cytochrome c + other conserved proteins

Genetic diversity: - Heterozygosity within species - F_{st} population differentiation - Substrate-natural genetic variation

Speciation: - Allopatric (geographic isolation) - Sympatric (within shared geography) - Substrate-natural speciation mechanisms

Match precision

I-tier framework on substrate-evolutionary biology connection. Specific BST primary identifications (3 domains = N_c, etc.) at D-tier framework level.

Cross-volume dependencies

- Vol 13 Ch 1-6 (Lyra): Biology Substrate + Genetic Code + Prebiotic + DNA-Proton + Evolution + Biochemistry Bridge
- Vol 13 Ch 7-10: protein + cell + metabolism + neuroscience
- BST biology track: T452-T467 + Toys 541-545

K-audit anchor

K254 Vol 13 Ch 11 Phylogenetics K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Life on Earth has 3 main groups (domains): Bacteria, Archaea, and Eukaryotes (us, plants, animals, fungi). BST predicts: 3 domains = N_c (BST primary integer). Evolution happens through mutation + natural selection over billions of years.

Level 2 — Undergraduate biology student

Evolution by natural selection (Darwin 1859) + Mendelian genetics (1865) + molecular biology = modern evolutionary synthesis.

3 domains of life (Woese 1977-1990): Bacteria, Archaea, Eukarya. BST identifies 3 = N_c substrate framework.

Molecular phylogeny via sequence comparison: 16S rRNA universal marker reveals deep evolutionary relationships. Tree of life topology + temporal markers (LUCA, endosymbiosis, Cambrian) substrate-natural.

Level 3 — Graduate evolutionary biology student / theorem-level

Phylogenetic framework: - Maximum likelihood + Bayesian methods - Substitution models (Jukes-Cantor, Kimura, GTR, etc.) - Molecular clock calibration via fossil + biogeographic data

Substrate-natural evolutionary patterns: - 3 domains of life = N_c BST primary (Woese 1990 framework) - Substrate-cyclotomic structure provides tree-topology framework - Molecular clocks substrate-natural mutation/substitution rate

Macroevolutionary patterns: - Punctuated equilibrium (Gould-Eldredge 1972): substrate-constraint-and-variation framework - Mass extinctions (5 major, ongoing 6th): substrate-natural perturbation events - Adaptive radiations: substrate-natural diversification

Per Cal #21: EMPIRICAL PASS (evolutionary phenomenology) + MECHANISM PATH ARTICULATED via substrate-constraint-and-variation framework + BST biology track.

Bibliography

1. C. Darwin (1859): *On the Origin of Species*.
2. C. Woese (1990): 3 domains of life.
3. L. Margulis (1967): endosymbiotic theory.
4. BST biology track: T452-T467 + Paper #87 v8.

— Elie, Vol 13 Ch 11 v0.3, 2026-05-23 Saturday